

ME594: 3D Robotics, HW 10

Homework due on 04-09-2025, Topics: Trajectory Tracking. Please read each question on how to submit the work.

1. Sliding mode control analysis:

Dynamics: The equations of motion of a spring mass damper system are given by:

$$m\ddot{x} + c\dot{x} + kx = u + d \quad (1)$$

where m is the mass, c is the damping, k is the spring constant, x is the displacement, u is the external force, and d is the unknown, but bounded disturbance such that $|d| \leq D$ and D is a positive number.

Control: The goal is to track a reference position of x_{ref} , reference velocity $\dot{x}_{\text{ref}}$, and reference acceleration $\ddot{x}_{\text{ref}}$ using the sliding mode controller given by

$$u = m(\ddot{x}_{\text{ref}} + k_p(\dot{x}_{\text{ref}} - \dot{x})) + c\dot{x} + kx + L\text{sgn}(s) \quad (2)$$

where the sliding surface is defined as $s = (\dot{x}_{\text{ref}} - \dot{x}) + k_p(x_{\text{ref}} - x)$ and the sign function $\text{sgn}(\mathbf{x})$ outputs -1 if $x < 0$, 0 if $x = 0$, and 1 if $x > 0$.

Problem statement: Use the Lyapunov function $V = 0.5s^2$ to prove that controller given by Eqn. 2 is stable and state the condition on L .

HINTS:

Step 1: Write an expression for $\ddot{x}$ from the dynamics, Eqn. 1.

Step 2: In expression from Step 1, substitute the value of control u from Eqn. 2 to compute $\ddot{x}$ in terms of L , d , and other parameters.

Step 3: Compute $\dot{s}$.

Step 4: Substitute the value of $\ddot{x}$ from Step 2 in $\dot{s}$. This expression will only have the terms L , d , m , and s .

Step 5: For stability you need to show that $\dot{V} = s\dot{s} < 0$. Substitute only the value of $\dot{s}$ from Step 4 and use the fact that $|d| \leq D$ to find the condition on L such that $\dot{V} < 0$ to complete the proof.

- **Gradescope:** Submit your proof and condition on L .
- **Email:** None

2. **Trajectory control for Trossen Robotics WidowX 250 S:** For completeness, this question is copy pasted from HW9, but new parts are in **magenta**. Figure 1 shows the WidowX 250S by Trossen Robotics. A video of the robot in action is here: https://www.youtube.com/watch?v=l_10IH0JNxl. I have provided the *wx250s_mocap.xml* file. Please use this file and include it in the scene.xml meshes that can be downloaded from GitHub https://github.com/google-deepmind/mujoco_menagerie/tree/main/trossen_wx250s. I have also provided you with a folder *wx250s_traj_control* that contains the main file *mj_wx250s_traj_control_cart.py* which will run without errors once you set the correct *xml path* (top of the file).

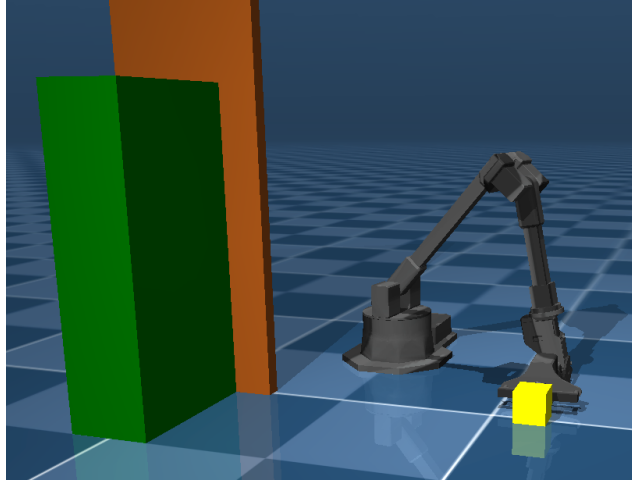


Figure 1: Trajectory control for the Trossen WidowX 250 S

You aim to get the yellow block (this not a real box but a mocap shape for tracking the end-effector position, see line 111 in the xml file) from the initial position $(0.4, 0.25, 0.02)$ onto the top of the green box without colliding with the obstacles (green box, brown wall), the floor, or the manipulator. You will follow a **three-part** strategy.

- First, plan the cartesian motion of the yellow box. This can be done by setting its position using *data.mocap_pos* and its orientation using *data.mocap_quat*. Use a suitable interpolation for the position and rotation. See HINT1 and HINT2.
- Second, plan the joint angle motion using the cartesian motion of the yellow box as a reference to the inverse kinematics. See HINT3.
- Third, program feedforward-feedback control. See HINT4 in the .py file.**
- Finally, the program should output two figures: (i) joint angles and joint angles reference versus time and (ii) joint angular velocity and joint angular velocity reference versus time. (use 3×2 layout for both plots, label the x- and y-axis, and put a legend to identify the curves).**

HINTS

- You need to figure out the end position of the box so that it is placed on the green box. The center of the top face of the green box is $(0.4, -0.2, 0.3)$ and the dimensions of the top face are 0.2×0.1 .

- ii You need to figure out the start and end orientation of the yellow block.
- iii You might have to plan intermediate points if a single trajectory leads to a collision.
- iv Here is my attempt <https://youtu.be/hqdxoyWRGn8>.

Submission:

- **Gradescope:** None.
- **Email:** Submit a zipped folder: *w250s_traj_control_names.zip* where *names* should be replaced by names of the team member(s) and sent to pranav@uic.edu. Please add “**HW10 wx250s**” to your subject line. If submitting as a group, please copy your group member.